MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

FIFTH SEMESTER

ELECTIVE – MG5DSEBTG300 - BIOTECHNOLOGY AND ENTREPRENEURSHIP

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word answer

Answer any **10** questions.

Each question carries **2** marks

1. Define Bioentrepreneurship. [U] / CO1
2. Name any two essential traits of a successful bioentrepreneur. [K] / CO1
3. What is the significance of bioentrepreneurship in the modern business landscape? [U] / CO1
4. Give one example of a contribution of biotechnology to economic growth. [K] / CO1
5. Define Intellectual Property Rights. [U] / CO2
6. What is a patent? [U] / CO2
7. Name the act that governs trademarks in India. [K] / CO2
8. What is the term of protection for copyright in India? [K] / CO2
9. Name any one major funding agency supporting bioentrepreneurs in India. [K] / CO3
10. Mention one policy initiative of the Indian Government for biotechnology. [K] / CO3
11. Give an example of a global partnership in biotechnology. [K] / CO3
12. Name one major area of medical biotechnology entrepreneurship. [K] / CO4

(10 × 2 = 20)

Part B

Short essay questions

Answer any **4** questions.

Each question carries **5** marks

13. Explain the link between biotechnology and entrepreneurship. [U] / CO1
14. Describe the essential bio entrepreneurial characteristics required for success. [U] / CO1
15. Discuss the legal and ethical implications in bioentrepreneurship. [E] / CO2
16. Explain the importance of branding and marketing strategies in bioentrepreneurship. [E] / CO2

17. Write notes on the role of the Indian Government in promoting bioentrepreneurship. [A] / CO3
18. Briefly explain entrepreneurship opportunities in environmental biotechnology. [E] / CO4

(4 × 5 = 20)

Part C

Essay questions
Answer any **2** questions.
Each question carries **15** marks

19. Discuss in detail the process of ideation, conceptualization, and research & development in bioentrepreneurship. How does resilience and adaptability contribute to small-scale product development? [U] / CO1
20. Write a detailed account on Intellectual Property Rights with special reference to patents, trademarks and copyrights. Evaluate their significance in protecting bio entrepreneurial innovations. [E] / CO2
21. Explain the various policies, initiatives, funding opportunities and grants available for bioentrepreneurs in India. How do foreign investments and global collaborations support the growth of biotechnology in India? [A] / CO3
22. Write an essay on specialized areas of bioentrepreneurship covering agricultural, medical, industrial and environmental biotechnology. Also discuss the global outlook of bioentrepreneurship. [E] / CO4

(2 × 15 = 30)